

D613

Decision Intelligence

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
-----------------------------	---	------------------------	----------------------------------

Items

1.

A logistics firm's routing team picked a route using the best forecast available, which gave it a 90% chance of on-time arrival. A rare rockslide closed the road and the shipment arrived two days late. The team lead has 15 years of experience, and the client is a national grocery chain. How should the routing decision be judged?

- A. As a sound decision, because the lead has 15 years of experience.
- B. As a poor decision, because the shipment was late.
- C. As a poor decision, because routing should always guarantee on-time arrival.
- D. As a sound decision that met a bad outcome.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A bank's machine-learning model scores small-business loan applications. For 85% of applications it is highly confident; the rest fall in a gray zone near the approval cutoff. The applications run through a modern cloud data platform. Where should a human-in-the-loop design place the loan officers?

- A. Nowhere; the model is more consistent than people.
- B. Monitoring the cloud platform's uptime.
- C. Reviewing every application by hand.
- D. Reviewing the gray-zone applications before a final decision.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A subscription company draws a causal decision diagram for the goal "reduce customer churn." It lists monthly price (set by the company), competitor promotions, customer satisfaction score, and churn rate. The diagram came out of a two-hour workshop with eight people. Which element is a lever?

- A. Monthly price.
- B. Competitor promotions.
- C. Customer satisfaction score.
- D. Churn rate.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A team scores three vendors from 1 to 10. Weights are cost 0.5, quality 0.3, support 0.2. Vendor A scores 10, 6, 4; Vendor B scores 6, 9, 9; Vendor C scores 7, 8, 6. Vendor B's sales representative used to work at the company. Which vendor wins the weighted scoring?

- A. Vendor B, because the team already knows the representative.
- B. Vendor C, because it has no score below 6.
- C. Vendor A, with 7.6.
- D. Vendor B, with a total of 24.

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
-----------------------------	---	------------------------	----------------------------------

5. [TRANSFER — reversed stem / different variable]

With weights of cost 0.5, quality 0.3, and support 0.2, Vendor D scores 8 on cost and 6 on quality. What support score does D need to reach a weighted total of 7.0?

- A. 6
- B. 7
- C. 1.2
- D. 0.24

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
-----------------------------	---	------------------------	----------------------------------

6.

A retailer can run a promotion that costs \$50,000. There is a 60% chance it adds \$150,000 in margin and a 40% chance it adds \$20,000. The marketing VP is sure it will be a hit. What is the expected net value of running the promotion?

- A. \$98,000
- B. \$48,000
- C. \$100,000
- D. \$35,000

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
-----------------------------	---	------------------------	----------------------------------

7. [TRANSFER — reversed stem / different variable]

A project costs \$30,000. If it succeeds it returns \$100,000; if it fails it returns nothing. What probability of success makes the expected net value exactly zero?

- A.** 70%
- B.** 50%
- C.** 30%
- D.** About 23%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A 600-bed hospital has spent \$4 million on a scheduling system that testing shows will not meet its needs. Finishing it would cost \$3 million more; a proven alternative costs \$2 million. One board member says, “We can’t waste the \$4 million we’ve already spent.” Which bias is at work?

- A.** No bias; protecting prior investment is prudent.
- B.** Anchoring.
- C.** Sunk cost fallacy.
- D.** Confirmation bias.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

Which situation shows anchoring?

- A.** An analyst reads only reports that support her forecast.
- B.** A buyer’s counteroffer stays close to the seller’s high opening price even after learning the item is worth far less.
- C.** A manager keeps funding a failing project because of money already spent.
- D.** A traveler overestimates plane-crash risk after seeing news coverage.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A distributor installed a decision-intelligence tool for inventory reorders. First-year cost was \$200,000. It cut stockouts and carrying costs, saving \$260,000 over the year. The vendor's dashboard has 40 charts. What is first-year ROI?

- A. \$60,000
- B. About 23%
- C. 130%
- D. 30%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

A decision tool must show at least 25% first-year ROI. It is expected to save \$150,000. What is the most it can cost?

- A. \$187,500
- B. \$120,000
- C. \$112,500
- D. \$37,500

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A claims department is moving from manager judgment to model-recommended claim routing. Many adjusters do not know why the change is happening, and rumors say the model will replace them. The new system has a sleek mobile app. What should change management do first?

- A. Explain why the change is needed and how adjusters' roles will and will not change.
- B. Mandate use and track compliance weekly.
- C. Start hands-on training on the new routing tool.
- D. Promote the mobile app's features.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A logistics firm's routing team picked a route using the best forecast available, which gave it a 90% chance of on-time arrival. A rare rockslide closed the road and the shipment arrived two days late. ~~The team lead has 15 years of experience, and the client is a national grocery chain.~~ How should the routing decision be judged?

Target word: judging decision quality **Decides:** The choice used the best information available at the time.

A	Camouflage	Right label, wrong reason; experience does not decide decision quality.
B	Inverted operation	Judges the decision by its outcome (resulting).
C	Absolute language	No decision under uncertainty can guarantee an outcome.
D	Correct	Decision quality rests on the process and information at the time, not luck afterward.

2. Answer D Concept 2

A bank's machine-learning model scores small-business loan applications. For 85% of applications it is highly confident; the rest fall in a gray zone near the approval cutoff. ~~The applications run through a modern cloud data platform.~~ Where should a human-in-the-loop design place the loan officers?

Target word: where humans review **Decides:** Low-confidence gray-zone cases carry the most risk of error.

A	Absolute language	Removes human oversight where it matters most.
B	Camouflage	The platform detail does not decide where judgment belongs.
C	Adjacent category	That is manual processing, not augmentation; it throws away the model's value.
D	Correct	Human judgment is spent where the model is least sure.

3. Answer A Concept 3

A subscription company draws a causal decision diagram for the goal "reduce customer churn." It lists monthly price (set by the company), competitor promotions, customer satisfaction score, and churn rate. ~~The diagram came out of a two-hour workshop with eight people.~~ Which element is a lever?

Target word: identify the lever **Decides:** Only monthly price is directly set by the company.

A	Correct	A lever is an action the decision maker controls.
B	Confusion set	An external: it affects churn but the company cannot set it.
C	Adjacent category	An intermediate: it is moved by levers, not set directly.
D	Inverted operation	That is the outcome being measured.

4. Answer C Concept 4

A team scores three vendors from 1 to 10. Weights are cost 0.5, quality 0.3, support 0.2. Vendor A scores 10, 6, 4; Vendor B scores 6, 9, 9; Vendor C scores 7, 8, 6. ~~Vendor B's sales representative used to work at the company.~~ Which vendor wins the weighted scoring?

Target word: weighted winner **Decides:** $A = 5.0 + 1.8 + 0.8 = 7.6$; $B = 3.0 + 2.7 + 1.8 = 7.5$; $C = 3.5 + 2.4 + 1.2 = 7.1$.

A	Camouflage	The relationship is not a scored criterion.
B	Adjacent category	That is a different decision rule (avoid weak spots), not weighted scoring.
C	Correct	Highest weighted total.
D	Dropped element	Adds raw scores and drops the weights.

5. Answer A Concept 4 · transfer

With weights of cost 0.5, quality 0.3, and support 0.2, Vendor D scores 8 on cost and 6 on quality. What support score does D need to reach a weighted total of 7.0?

Target word: solve for a missing score **Decides:** $4.0 + 1.8 = 5.8$, so support must add 1.2.

A	Correct	$1.2 / 0.2 = 6$.
B	Wrong stem number	The target total, not the score needed.
C	Dropped element	The weighted points needed, before dividing by the weight.
D	Inverted operation	Multiplies 1.2 by the weight instead of dividing.

6. Answer B Concept 5

A retailer can run a promotion that costs \$50,000. There is a 60% chance it adds \$150,000 in margin and a 40% chance it adds \$20,000. ~~The marketing VP is sure it will be a hit.~~ What is the expected net value of running the promotion?

Target word: expected net value **Decides:** $0.6 \times 150,000 + 0.4 \times 20,000 = 98,000$ before cost.

A	Dropped element	Leaves out the \$50,000 cost.
B	Correct	$\$98,000$ expected margin – $\$50,000$ cost = $\$48,000$.
C	Camouflage	Assumes the hit the VP expects ($\$150,000 - \$50,000$) and ignores the odds.
D	Adjacent category	Uses a plain average of outcomes ($\$85,000 - \$50,000$), not probability weights.

7. Answer C Concept 5 · transfer

A project costs \$30,000. If it succeeds it returns \$100,000; if it fails it returns nothing. What probability of success makes the expected net value exactly zero?

Target word: solve for break-even probability **Decides:** $p \times 100,000 - 30,000 = 0$.

A	Inverted operation	That is the failure probability at break-even.
B	Absolute language	Assumes every bet breaks even at a coin flip.
C	Correct	$p = 30,000 / 100,000 = 0.30$.
D	Confusion set	$30,000 / 130,000$ adds cost to payoff; payoff already is the gross return.

8. Answer C Concept 6

~~A 600-bed~~ hospital has spent \$4 million on a scheduling system that testing shows will not meet its needs. Finishing it would cost \$3 million more; a proven alternative costs \$2 million. One board member says, “We can’t waste the \$4 million we’ve already spent.” Which bias is at work?

Target word: name the bias **Decides:** Past, unrecoverable spending is driving a future choice.

A	Inverted operation	Finishing costs \$3 million for a worse result than the \$2 million alternative.
B	Confusion set	Anchoring is sticking near a first number offered, not honoring past spending.
C	Correct	The \$4 million is gone either way; only future costs and benefits should count.
D	Adjacent category	Confirmation is seeking agreeing evidence; the board member is citing past cost, not filtering evidence.

9. Answer B Concept 6 · transfer

Which situation shows anchoring?

Target word: bias — reversed from name to example **Decides:** A first number pulls later estimates toward it.

A	Adjacent category	That is confirmation bias.
B	Correct	The first number anchors the estimate.
C	Confusion set	That is sunk cost.
D	Confusion set	That is availability.

10. Answer D Concept 7

A distributor installed a decision-intelligence tool for inventory reorders. First-year cost was \$200,000. It cut stockouts and carrying costs, saving \$260,000 over the year. ~~The vendor's dashboard has 40 charts.~~ What is first-year ROI?

Target word: ROI percentage **Decides:** Net gain = \$60,000 on a \$200,000 cost.

A	Adjacent category	Net benefit in dollars, not a return rate.
B	Inverted operation	$\$60,000 / \$260,000$ divides by benefit instead of cost.
C	Dropped element	$\$260,000 / \$200,000$ drops the subtraction of cost.
D	Correct	$(\$260,000 - \$200,000) / \$200,000 = 0.30$.

11. Answer B Concept 7 · transfer

A decision tool must show at least 25% first-year ROI. It is expected to save \$150,000. What is the most it can cost?

Target word: solve for maximum cost **Decides:** Savings = cost $\times$ 1.25.

A	Confusion set	$\$150,000 \times 1.25$ multiplies where it should divide.
B	Correct	$\$150,000 / 1.25 = \$120,000$; check: $\$30,000 / \$120,000 = 25\%$.
C	Inverted operation	$\$150,000 \times 0.75$ takes 25% off savings; ROI is measured on cost.
D	Dropped element	25% of savings, not the allowed cost.

12. Answer A Concept 8

A claims department is moving from manager judgment to model-recommended claim routing. Many adjusters do not know why the change is happening, and rumors say the model will replace them. ~~The new system has a sleek mobile app.~~ What should change management do first?

Target word: first change-management step **Decides:** Staff lack awareness of why and fear the change.

A	Correct	Awareness and desire come before knowledge and ability.
B	Inverted operation	Reinforcement belongs at the end, not before people understand the change.
C	Dropped element	Skips awareness and desire; training resisted is training wasted.
D	Camouflage	The app does not address the why or the fear.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	7	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	3	
Confusion set	Every option is real; the test is exact role-matching.	6	
Camouflage	Answers a stem fact that was never the question.	5	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Decision quality vs. outcome	Judge a decision by the information, options, and reasoning available when it was made, not by how it turned out.	“Resulting” — calling a decision bad because the outcome was bad.	A well-reasoned choice followed by bad luck.
2	Human-in-the-loop design	Route low-confidence, high-stakes, or novel cases to a human reviewer and let the model handle clear-cut cases.	Either full automation or reviewing every case by hand.	A model confidence score and a gray zone near a cutoff.
3	Causal decision diagram elements	Levers are actions the decision maker controls; externals are outside factors; intermediates link levers to outcomes; outcomes are the goals measured.	Calling an outside factor or a metric a lever.	“Which element can the company directly set?”
4	Weighted multi-criteria scoring	Multiply each criterion score by its weight and sum; the highest weighted total wins.	Adding raw scores and ignoring weights.	A score table plus criterion weights that sum to 1.
5	Expected value	Expected value = $\sum(\text{probability} \times \text{payoff})$, minus any certain cost.	Using the best case, or a plain average of outcomes.	Two or more outcomes with percentages.
6	Biases and heuristics	Sunk cost = letting past spending drive future choices; anchoring = sticking near a first number; availability = judging likelihood by what is easy to recall; confirmation = seeking agreeing evidence.	Mixing up the names, or calling a bias prudence.	“We already spent...”, a first offer, or a vivid recent event.
7	ROI of a decision change	$\text{ROI} = (\text{benefit} - \text{cost}) / \text{cost}$.	Dividing benefit by cost, or dividing by benefit.	A cost and a savings figure for one period.
8	Change management for new decision methods	People need awareness of why the change is happening and a desire to support it before training and reinforcement work (ADKAR order).	Jumping straight to training or mandates.	Staff who do not know why the change is happening or fear it.

Course scope (public description): Decision Intelligence is a domain that optimizes decision-making by balancing technology, processes, and people. In this course students learn the core principles of Decision Intelligence, exploring the augmentation of decision processes with machine learning, comprehensive decision modeling, and the pivotal role of a “human-in-the-loop” design. Students will navigate decision theories and multi-criteria decision analysis, gaining insight into how biases and heuristics influence decision outcomes. The course emphasizes framing decisions using causal decision diagrams (CDD), implementing decision intelligence, evaluating the outcome using key performance indicators and determining the return on investment of the change, and using change management techniques to help the organization adapt to new decision making strategies.